CATERPILLAR®

BEST PRACTICE SERIES

Magnetic Plug Inspection Trays and Inspection Process

Maintenance and Repair | Application | Component Renewal (CRC) | Component Life Management | MARC Management

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1.0 Introduction

During the PM process magnetic plug inspections and changes can be sped up by creating labeled trays of clean plugs that will replace the plugs in the machine. This tray would then be sent back to the CM analyst for inspection and result recording. By having a common person do the inspections, ratings will be consistent between inspections and machines.

Recording pictures of the magnetic plugs, plug ratings, and component life can help the CM analyst track changes in the magnetic plug condition from PM to PM. Tracking and trending the information from magnetic plug inspections can be used to gain better understanding of component life and plan for a PCR.

2.0 Best Practice Description

Magnetic plug trays should be prepared that contain clean magnetic plugs in individual, labeled bins (Figure 1.) When a plug is removed from the machine it is replaced by a clean plug from the tray and the used plug is placed in the appropriately labeled bin.

Figure 1: Two examples of labeled magnetic plug trays.

Once all the plugs on a machine are replaced, the tray containing the used plugs should be sent to a CM analyst who will inspect and record the condition of all the plugs. Pictures should be taken and saved with the component life hours and plugs should be rated based on the attached document. Saving pictures with the component life hours allows the analyst to view how a certain plug looked at the last PM compared to the current PM.

A database or spread sheet should be developed to help track and trend the component life data, and store plug ratings and pictures. Information gathered during this process can be used to gain a better understanding of component life and plan for a PCR.

By having the same personal examine and rate the magnetic plugs, as opposed to different mechanics working on the machines, you assure that all plugs are rated on a similar scale between machines and between repairs. This helps develop reliable component life estimates as well as eliminates unnecessary downtime when a component is incorrectly rated as needing repair.

Once the plugs have been inspected and the results recorded, plugs that can be returned to service should be cleaned and put in a bin ready for the next machine PM.

NOTE: Changing the magnetic plugs as part of the pre-PM process rather than part of the PM process can help in planning process. If the machine has significant wear and a component needs to be changed it can be properly scheduled and operations won't expect the machine back before it will be available.

3.0 Implementation Steps

1. Consider the type and number of machines, frequency of magnetic plug inspections, service location (pre-PM and/or PM bay), turnaround time for tray inspection (time in lab to inspect and release tray), etc.

2. Decide on a tray design based on mine preference and fabricated the appropriate number of trays. Equip the trays with new/recycled magnetic plugs.

3. Train the PM personnel in the correct procedures.

4. Develop a database or spread sheet to be used for tracking and trending component life.

5. Assign and train a person responsible for receiving and analyzing the magnetic plug trays and tracking component life.

6. Assign carts to PM bays and/or pre-PM locations.

7. Implement and control the correct execution of the working procedures.

4.0 Benefits

- Reduces time the machine is in the shop for magnetic plug inspections.
- Common inspection ratings with only 1 analyst inspecting plugs rather than multiple mechanics.
- Provides information that can help in component life and PCR planning.
- Avoid unnecessary downtime from mis-rated magnetic plugs leading to unneeded component repairs.

5.0 Resources Required

- Labeled trays for magnetic plug sets.
- Personal assigned to examining and rating magnetic plugs and tracking component life.

6.0 Supporting Attachments / References

7.0 Related Best Practices

1206-2.15-1047 Fluid Analysis Sampling Cart